

PROJECT TITLE : AI FAQ ASSISTANT API

PROJECT DESIGN PHASE

1. SYSTEM ARCHITECTURE DESIGN

The AI FAQ Assistant follows a layered backend architecture using Node.js and Express.js. The main components include the Express server, authentication middleware, controllers, Mongoose database layer, MongoDB database, and Google Gemini AI integration. The architecture processes user requests securely and connects them to the database or AI service as required.

2. DATABASE DESIGN

MongoDB is used as the primary database, with Mongoose for schema definition, validation, and database operations. The system contains User, FAQ, AI Generation, and Category entities. The User collection stores user details such as name, email, password, role, and timestamps. The FAQ collection stores questions, answers, categories, creator information, and timestamps.

3. AUTHENTICATION AND SECURITY DESIGN

The system uses JWT-based authentication to secure protected APIs. Passwords are encrypted using bcrypt before being stored in the database. Authentication middleware verifies the JWT token and controls access to protected operations. Different roles such as Admin, Content Creator, Authenticated User, and Public User are provided with different permissions.

4. API DESIGN

The application provides REST API endpoints for user registration, login, FAQ management, FAQ search, and AI-based FAQ generation. Private APIs are protected using JWT Bearer-token authentication. The APIs allow users to securely create, manage, search, and generate FAQ content.

5. AI INTEGRATION DESIGN

Google Gemini AI is integrated into the system to generate FAQ questions and answers. The AI generation feature works with the backend API and provides AI-assisted content generation for users.

6. ERROR HANDLING AND VALIDATION

The system includes centralized error handling and Mongoose schema validation. Validation ensures that appropriate data is stored in MongoDB, while centralized error handling provides reliable responses when errors occur.

